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EXPERIMENT 2

Exp No:2 Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

AIM:

To Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

DESCRIPTION:

Use sample data set sales-data.csv.

PROGRAM:

```
 import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Load the data into a pandas DataFrame
file_path = '/content/drive/MyDrive/Colab Notebooks/SALES DATA.csv'
df = pd.read_csv(file_path)

# Display the first few rows of the DataFrame
print(df.head())

# Check for missing values
print(df.isnull().sum())

# Fill missing Sales values with the mean
df['Sales'] = df['Sales'].fillna(df['Sales'].mean())

# Drop rows with missing important values
df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)

# Summary statistics
print(df.describe())

# Group by product and calculate the total sales and quantity
product_summary = df.groupby('Product').agg({
    'Sales': 'sum',
    'Quantity': 'sum'
}).reset_index()

print(product_summary)

# Bar plot of total sales by product
plt.figure(figsize=(10, 6))
```

```
plt.bar(product_summary['Product'], product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.show()

# Line plot of sales over time
df['Date'] = pd.to_datetime(df['Date'], dayfirst=True)

sales_over_time = df.groupby('Date').agg({
    'Sales': 'sum'
}).reset_index()

plt.figure(figsize=(10, 6))
plt.plot(sales_over_time['Date'], sales_over_time['Sales'])
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('Sales Over Time')
plt.show()

# Pivot table to analyze sales by region and product
pivot_table = df.pivot_table(
    values='Sales',
    index='Region',
    columns='Product',
    aggfunc='sum',
    fill_value=0
)

print(pivot_table)

# Correlation matrix
correlation_matrix = df[['Sales', 'Quantity']].corr()

print(correlation_matrix)

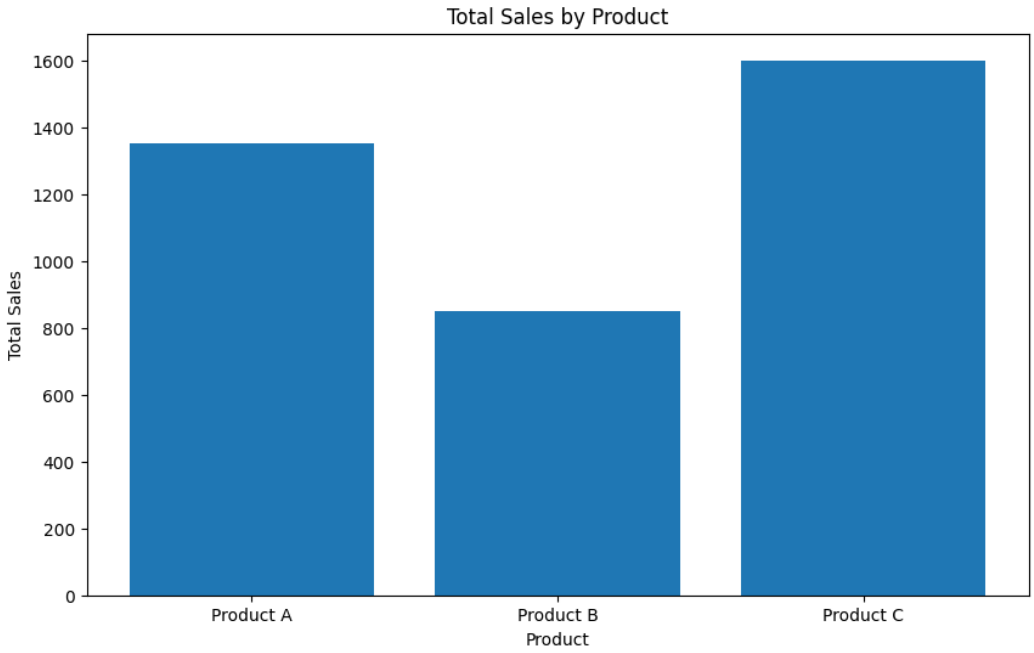
# Heatmap of the correlation matrix
import seaborn as sns

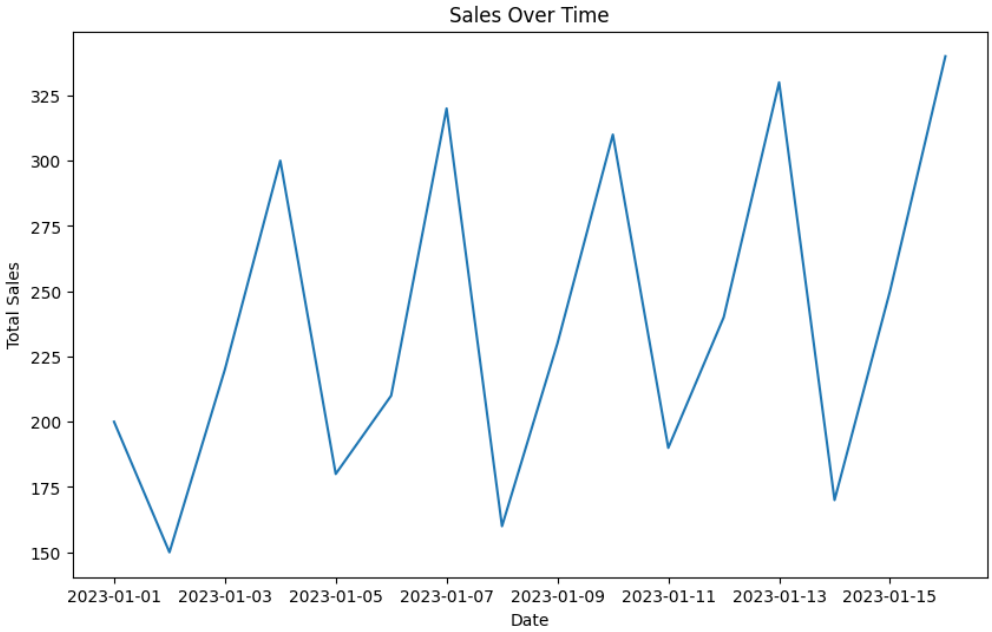
plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')

plt.title('Correlation Matrix')
plt.show()
```

OUTPUT:

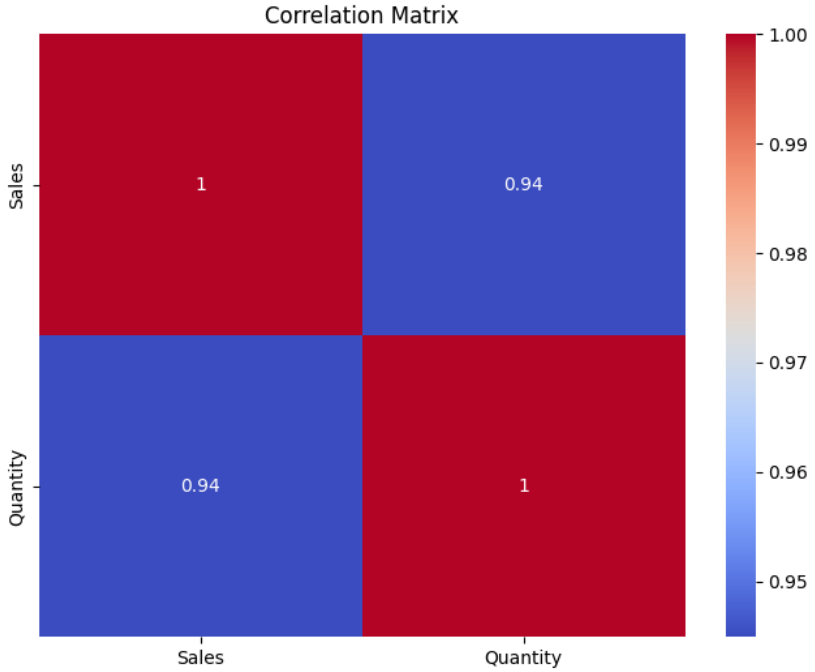
```
    Date    Product  Sales  Quantity Region
0  01-01-2023  Product A    200         4  North
1  02-01-2023  Product B    150         3  South
2  03-01-2023  Product A    220         5  North
3  04-01-2023  Product C    300         6   East
4  05-01-2023  Product B    180         4   West
Date        0
Product     0
Sales       0
Quantity    0
Region      0
dtype: int64
            Sales   Quantity
count   16.000000  16.000000
mean   237.500000   5.375000
std     64.031242   1.746425
min    150.000000   3.000000
25%    187.500000   4.000000
50%    225.000000   5.500000
75%    302.500000   7.000000
max    340.000000   8.000000
     Product  Sales  Quantity
0  Product A   1350        33
1  Product B    850        17
2  Product C   1600        36
```





Product	Product A	Product B	Product C
Region			
East	0	0	1600
North	1350	0	0
South	0	480	0
West	0	370	0

	Sales	Quantity
Sales	1.000000	0.944922
Quantity	0.944922	1.000000



RESULT:

Thus the given CSV dataset was successfully uploaded, analyzed and preprocessed. The data was visualized using different plots and the required information was acquired.